Student 4 **Chintha Abhilash Reddy** 192525462RC 5 / 5 submitted

Q1. Harshad number

ANSWER

```
n = int(input())

s = sum(int(i)  for  i in str(n))

if n % s == 0:
    print("harshad number")
else:
   print("not a harshad number")

print("harshad numbers from 1 to 500:")
for n in range(1,501):
     if  n % sum(int(i) for i in str(n)) == 0:
           print(n,end=" ")
```

Q2. GCD

CODE

```
a = int(input("Enter the first:\n"))
b = int(input("Enter the secound:\n"))

while b ≠ 0:
    temp = b
    b = a % b
    a = temp

print("Gcd is",a)
```

INPUT

```
24
36
```

OUTPUT

```
Enter the first:
Enter the secound:
Gcd is 12
```

Q3. sum to a target value

ANSWER

```
def find_pairs(lst,target):
seen = set()
pairs = []
 from num in lst:
      complement  = target - num

if complement in seen:
        pairs.append((complement,num))

  seen.add(num)

return pairs

data = [int(x) for x in input("Enter numbers: ").split()]
t = int(input("Target: "))
result = find_pairs(data,t)
print("pairs:",result)
```

Q4. Bitwise Operators

ANSWER

```
a = 13
b = 21
c = 26
result = 31
0b11111
0x1f
7 62 -32
```


Q5. binary search

CODE

```
def rotated_binary_search(arr,target):
    low = 0
```

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Q5. binary search

CODE

```
def rotated_binary_search(arr,target):
    low = 0
    high = len(arr)-1

    while low ≤ high:
        mid = (low + high)// 2

        if arr[mid] == target:
            return mid

        if arr[low] ≤ arr[mid]:
            if arr[low] ≤ target < arr[mid]:
                high = mid - 1
            else:
                low = mid + 1

        else:
            if arr[mid] < target ≤ arr[high]:
                    low = mid + 1
            else:
                high = mid - 1

    return -1
data = [4,5,6,7,0,1,2]
target = int(input("Target: "))

result = rotated_binary_search(data,target)
print("Found at:",result)
```

OUTPUT

```
Target: Found at: 4
```